Aarav Jain

Undergraduate Researcher, Computer Science

Tampa, FL | 813-525-4765 | aaravj651@gmail.com
aadraxe.github.io | github.com/aadraxe | LinkedIn

Research Interests

Human-robot interaction; socially assistive robotics; robot learning; explainable AI and robot intent explanation; trustworthy and secure LLM agents; authorization and information-flow control for agents.

Education

University of South Florida, Tampa, FL — **B.S. Computer Science**
Judy Genshaft Honors College — 2023 – May 2027 (expected)

- GPA: 3.95/4.00
- Relevant coursework: Machine Learning, Data Structures and Algorithms, Operating Systems, Computer Organization, Databases, Software Engineering

Publications

Workshop Papers

[W1] A. Ergogo, T. A. Nguyen, M. Anjoul, **A. Jain**, Y. Zaheen, D. Shrestha, and Z. Han. "Towards Contextual Robot Intent Explanation for Hierarchical Vision-Language-Action Collaboration." In *ICRA 2026 Workshop on Semantics for Reliable Robot Autonomy: From Environment Understanding and Reasoning to Safe Interaction*, 2026. **Accepted.**

Under Review

[R1] R. Phatak*, H. Wang*, **A. Jain**, N. Echeverria, P. S. Bhele, and Z. Han. "Proactive Robot Abuse Mitigation via Physical Protectives." Under review at *International Journal of Social Robotics*, 2026. (*Equal contribution.)

[R2] X. Kong*, **A. Jain***, T. A. Nguyen, and Z. Han. "Can We Use Augmented Reality Robots for HRI User Studies? An Investigation in Mobile Manipulation Context." In revision for *Int. J. of Social Robotics*, 2026. (*Equal contribution.)

Technical Reports

[T1] Co-author. "Threat Advisory: Vercel Security Incident." Cyber Florida, 2026.

[T2] Co-author. "Threat Advisory: RDP Exploitation and AI Token Theft." Cyber Florida, 2026.

Research Experience

RARE Lab, University of South Florida — **Undergraduate Researcher**
Advisor: Prof. Zhao Han — Feb 2025 – Aug 2026

- *Projects:* Four human-robot interaction projects across three robot platforms (Bloom, Fetch, Pepper), resulting in three papers, one as co-first author [W1, R1, R2].
- **Bloom platform (companion robot for older adults):** Built the on-device interaction stack: wake-word speech recognition (Vosk), LLM dialogue with text-to-speech and configurable pauses, real-time YOLO detection on a Raspberry Pi AI HAT+, and a live Flask control dashboard; cut voice-pipeline startup from 20–25 s to 5–10 s with lazy loading.
- **CRIE research [W1]:** Extended Bloom into CRIE's proactive communication layer. Built the task-detection component that separates task requests from casual conversation (89% accuracy versus 52% for a keyword baseline in an early test), plus camera capture and a two-way gRPC interface that sends task goals and workspace observations to the vision-language planner and returns its next steps.
- **AR vs. physical robots [R2]:** Programmed the Fetch mobile manipulator in ROS 2 and fixed the AR virtual-robot setup (HoloLens 2) for a 33-participant study comparing physical and virtual robots on accuracy, reaction time, usability, and trust; helped run sessions. Leading the journal revision as co-first author.
- **Proactive robot abuse mitigation:** Contributed to conceptualization, methodology, software, data curation, visualization, and the original draft [R1]; reprogrammed Pepper for the revised protocol and ran sessions (N = 32).
- *Responsibilities:* Robot software for all four projects; ran two IRB-approved counterbalanced within-subjects studies (N = 33, N = 32); co-wrote a journal draft [R1]; presented twice to faculty and funders.

Presentations

- "Bloom: A Companion Robot." Demonstration to University of South Florida faculty and funders.

- "Proactive Robot Abuse Mitigation via Physical Protectives." Presentation to University of South Florida faculty and funders, June 2025.

Honors, Awards, and Certifications

Presidential Green and Gold Scholarship, University of South Florida
UR2PhD REU Funding Award ($8,000), Computing Research Association — Fall 2025 – Summer 2026
Dean's List, University of South Florida (4 semesters)
3rd Place of 81 teams, Sperry Tech Challenge, ShellHacks 2026 — Sep 2026
Selected as Undergraduate Teaching Assistant, Computer Organization, USF
Blue Team Level 1 (BTL1), Security Blue Team — Mar 2026
UR2PhD Undergraduate Research Methods Course, Computing Research Association — Summer 2025

Professional Experience

Cyber Florida at USF, Tampa, FL — **Cybersecurity Analyst**
Security Operations Center — Jan 2026 – Present

- Triage up to 25 Microsoft Defender alerts a week; confirmed command-and-control traffic that led to 20+ malicious IPs and domains being blocked.
- Identified a Dockerfile enumeration campaign in web server logs and escalated a potential internal host compromise.
- Reconstructed 6 simulated attacks end to end in Splunk and CrowdStrike, mapping each step to MITRE ATT&CK; co-authored two published threat advisories [T1, T2].

HGrid, Remote (Tampa, FL) — **Software Engineer Intern**
Healthcare financing startup — May 2025 – Aug 2025

- Automated patient financing agreements for hospitals, from PDF extraction to auto-filled agreements and DocuSign e-signature, cutting a 3–4 hour manual task to about 20 seconds for 6+ hospitals; added alerts and a fallback so every agreement was delivered despite about 20% DocuSign failures.
- Integrated a credit bureau API so each patient's interest rate is set automatically from their credit score.

Leadership and Service

Senator, Student Government, University of South Florida — May 2025 – Dec 2025
Helped oversee a student budget of more than $20 million and voted on campus legislation.

External Affairs and Relations Lead, ACM Student Chapter, University of South Florida — Jul 2024 – Feb 2025
Brought in two industry speakers for the chapter's campus AI convention.

Selected Projects

- **AegisFlow: LLM and Agent Security** (Python, PyTorch, FastAPI, MCP, C#; github.com/aadraxe/LLM-middleware-security). Grew from an LLM-middleware summer project my Cyber Florida manager suggested into an authorization layer for AI agents in which every MCP tool call and final answer must pass default-deny policy checks, independent of the model's own judgment. Provenance tracking blocked 7/7 prohibited actions versus 5/7 with role checks alone; sensitivity labels on retrieved RAG documents gave 0 leaks across a 23-scenario attack suite. With a live gpt-oss-120b agent: 0 unauthorized tool executions at 0.8 ms enforcement overhead. A fine-tuned DistilBERT injection classifier reached 0.93 F1 in-distribution versus 0.76 for an LLM judge, but only 22% recall out of distribution, which motivated enforcement outside the model.
- **ContractMap** (FastAPI, MongoDB, Gemini, scikit-learn, GitHub Actions). Backend for a construction-coordination platform with Gemini project extraction and an approval-gated function-calling assistant; cost model with 27% median error versus 74% for the grid operator's estimate; added a DevSecOps pipeline (Gitleaks, Semgrep, pip-audit) in GitHub Actions that caught committed keys and patched 2 high-severity CVEs. 3rd of 81 teams, ShellHacks 2026.

Technical Skills

Languages and backend: Python, C, C++, C#, Java, JavaScript, SQL, FastAPI, ASP.NET Core, Django, Flask, REST APIs, gRPC, Docker, Git, Linux

AI and robotics: PyTorch, Hugging Face Transformers, LLM agents, MCP, tool calling, RAG, scikit-learn, YOLO, Vosk, ROS 2, Choregraphe

Security and data: Splunk, CrowdStrike, Microsoft Defender, MITRE ATT&CK, GitHub Actions, Semgrep, Gitleaks, pytest, xUnit, MongoDB, MySQL